

SPOT Rebuttal Results

Canonical refreshed experiments use seed 3495 and CLIP ViT-B/32 QuickGELU token-77 query features.

1. Base Configurations and Boundary Statistics

Dataset	Boundary configuration	Train avg.	Val avg.	Train max	Val max	Base Rsum
ACT	cosine; prom. -1.0 ; dist. 1; smooth 1	36.07	36.31	212	224	184.6
TVR	cosine; prom. 0.2; dist. 2; smooth 1	21.10	20.98	75	52	232.8
QVH	cosine; prom. 1.3; dist. 2; smooth 3; split-only	72.37	71.13	143	139	242.6

Common optimization: batch size 128, weight decay 0.01, warmup proportion 0.01, maximum 100 epochs, early stopping patience 10, hard negatives from epoch 5 with pool size 5, GT-timestamp Soft MIL, and hybrid segment reduction ($\alpha = 0.9$). ACT/TVR use segment:frame fusion 0.85 : 0.15; QVH uses 0.80 : 0.20.

2. Shared Boundary Hyperparameters

Hyperparameter	U1	U2
Method	multi-scale cosine	
Base window	8	
Multi-scale windows	{2, 4, 8, 16}	
Scale weights	{0.4, 0.3, 0.2, 0.1}	
Peak selection / normalization	prominence / MAD	
MAD prominence	0.4	0.2
Minimum distance / smoothing	1 / 1	
Postprocessing	none	

Dataset	Base Rsum	U1 Rsum	U2 Rsum
ACT	184.6	183.1	183.7
TVR	232.8	232.8	233.1
QVH	242.6	242.4	245.5
Charades	–	56.8	55.7

Charades’ separately tuned best run reports $R@1/R@5/R@10/R@100 = 1.4/6.4/11.2/43.9$ and $Rsum = 62.9$.

3. Intra-Query Segment Separation and QVH Length Regularization

Dataset	Setting	R@1	R@5	R@10	R@100	Rsum
ACT	base IQS max	15.2	36.5	49.8	83.1	184.6
ACT	IQS log-sum-exp	14.5	36.3	49.6	83.2	183.6
TVR	base IQS log-sum-exp	26.3	51.6	62.5	92.4	232.8
TVR	IQS max	26.9	51.7	62.3	92.0	233.0
QVH	base IQS log-sum-exp	25.5	53.8	66.9	96.4	242.6
QVH	IQS max	24.2	53.4	65.5	96.5	239.6
QVH	no split-only length regularization	25.8	55.2	66.0	96.1	243.0

4. Segment/Frame Fusion Sensitivity

Segment:Frame	ACT Rsum	TVR Rsum	QVH Rsum
0.60:0.40	181.8	228.6	238.3
0.65:0.35	182.7	229.9	239.9
0.70:0.30	183.3	231.2	241.4
0.75:0.25	183.9	232.1	241.5
0.80:0.20	184.7	232.6	242.6
0.85:0.15	184.6	232.8	243.3
0.90:0.10	184.5	232.8	243.2

QVH evaluated at the ACT/TVR fusion coefficient 0.85 : 0.15 obtains $R@1/R@5/R@10/R@100 = 26.3/53.7/66.8/96.5$ and $Rsum = 243.3$.

5. Earlier GT-Timestamp Partition Ablation (Seed 9527)

Partition	Objective	R@1	R@5	R@10	R@100	Rsum
SPOT	no timestamp	22.0	47.9	61.0	94.3	225.2
SPOT	Soft MIL	25.1	54.1	65.8	96.5	241.4
SPOT	Soft MIL + IQS	26.6	51.7	65.9	96.5	240.7
Uniform matched	no timestamp	21.9	50.0	63.0	94.9	229.8
Uniform matched	Soft MIL	25.7	55.0	66.8	96.1	243.7
Uniform matched	Soft MIL + IQS	2.4	7.9	12.5	54.1	76.9

6. Boundary Construction Wall-Clock Time

Dataset	Wall-clock time (s)
ACT36	29.981
TVR21	53.653
QVH72	47.766

Timing used the base boundary generation scripts with eight workers and includes source-index generation plus train/validation boundary generation.